

CSE427 Section 5 Makeup Quiz

Fall 2025

Duration: 30 Minutes

Name:

ID:

Question 1 [5 Points]

Prove that, $\tanh(x) = 2\sigma(2x) - 1$ [show step by step].

Question 2 [5 Points]

An input image of size $224 \times 224 \times 3$ is passed through a sequence of convolutional layers. In Phase 1, the input is convolved with 32 filters of size $7 \times 7 \times 3$, with stride = 2 and zero padding = 3. In Phase 2, the output from Phase 1 is convolved with 64 filters of size $5 \times 5 \times 32$, with stride = 1 and zero padding = 2. In phase 3, the output from Phase 2 is convolved with 128 filters of size $3 \times 3 \times 64$, with stride = 2 and zero padding = 1. Determine the output size after each phase.

Solution

Question 1

$$2\sigma(2x) - 1 = \frac{2}{1+e^{-2x}} - 1 = \frac{1-e^{-2x}}{1+e^{-2x}} = \frac{e^x(1-e^{-2x})}{e^x(1+e^{-2x})} = \frac{e^x-e^{-x}}{e^x+e^{-x}} = \tanh(x)$$

Question 2

After Phase 1, output size = $112 \times 112 \times 32$

After Phase 2, output size = $112 \times 112 \times 64$

After Phase 3, output size = $56 \times 56 \times 128$